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Engineering lead who owns product **end to end**: product research, UI design, frontend, backend and infra. Founding engineer twice (AutoFlow, NativeBridge). Builds **agentic AI products**, **real-time device streaming over WebRTC** and **developer tools**; ships products of his own on the side (**Chukta**, **Markdrop**), open-sources the tools and teaches system design to backend engineers.

EXPERIENCE

Senior Software Engineer & Engineering Lead · Senseloaf AI
Apr 2026 - Present · Bengaluru

- Lead engineering of an **agentic hiring platform** used by enterprises and startups, with AI agents for **sourcing, resume-to-job matching, profile verification, prescreening and AI interviews**.
- Building **SIA (Senseloaf Intelligent Agent)**, a chat interface where a recruiter runs hiring end to end from one conversation, and the **MCP server** behind it.
- Own features end to end: product research, UI design, frontend, backend and **infra scaling**.

Software Engineer (Freelance) · pCloudy
Mar 2026 - Jun 2026 · Remote

- Researched **device streaming** for a cloud of **5,000+** devices, and built mobile app testing solutions.

Founding Engineer · NativeBridge
Jan 2025 - Mar 2026 · Bengaluru

- Built cloud **Android and iOS devices streamed to the browser over WebRTC** at **<100 ms**, with remote touch, rotation, clipboard sync and file transfer.
- Created **iOS Bridge** (open source, **75★**, 10 forks): a **FastAPI** server on macOS that drives iOS simulators through **simctl and idb** and streams them to Electron desktop apps (Windows, Linux, macOS), a browser client and a pip-installable CLI. Sessions survive server restarts; app install, MP4 recording and location simulation over REST and WebSocket APIs.
- Replaced screenshot-over-WebSocket streaming (**2–5 FPS**, ~2 s lag) with a **WebRTC video track** fed from a one-frame buffer that always keeps the newest frame: **60–90 FPS** at **50–100 ms**.
- Built **BridgeLink** (open source, on PyPI and Homebrew): a Python CLI that exposes USB or Wi-Fi Android devices to the NativeBridge platform through **API-key-authenticated bore tunnels**, so a remote developer runs **adb** as if the phone were plugged in. Background daemons health-check every device **each second** and re-activate it when it reconnects.
- Built the cloud infrastructure for **concurrent streaming sessions** on AWS and GCP: automatic resource allocation, **container orchestration** and **connection pooling**.
- Led product architecture, feature priorities and the technical roadmap from **MVP to production**, and mentored team members.

Founding Engineer #1 · AutoFlow
Jan 2024 - Dec 2024 · Bengaluru

- Employee #1: built the **whole backend from an empty repo** in **FastAPI** on PostgreSQL, plus a **Bun WebSocket gateway**.
- Shipped **AI-driven autonomous UI testing**: LLM agents that write and run UI tests on a fleet of **Playwright runners**, with runner workers across **AWS and GCP**.
- Launched on Product Hunt as **Product of the Day**.

Software Development Engineer 1 · Ikarus 3D
Jul 2023 - Oct 2023 · Chandigarh

- Led design and development of **3 business-critical applications**: carts and orders, the authentication system and the FDM slicer.
- Migrated the online FDM slicer from **AWS ECS and EFS to GCP Cloud Run**, solving runtime vertical scaling and cutting its cost by **90%**; designed it as a **serverless** service on Cloud Run, AWS Lambda and EFS.
- Wrote **Go** utilities for container lifecycle automation and persistent state storage, and added **noVNC** desktop streaming for the slicer applications.
- Set up the company's **first CI/CD pipelines** for business applications: faster deploys, fewer build failures and integration issues.
- Extended authentication with **device-ID tracking**, new sign-in notifications, password recovery and email change.
- Researched **3D generative AI** with Dr. Sahil Sharma, turning text and 2D images into 3D models with NeRF, BLIP and DreamFusion: **48%** accuracy within 30 days.
- Automated the company's **HRMS** with biometric hardware integration over APIs and webhooks, saving **132 hours** a year of HR work; mentored **3 interns**.

Software Developer Associate · Ikarus 3D
Jan 2023 - Jun 2023 · Chandigarh

- Led key backend microservices for the launch of **IkarusNest**: authentication, carts, orders and slicer.
- Built email/password and social login with **JWT sessions**, cutting login time by **30%**.

- Added **JWT header-based service-to-service auth** and caching for REST calls, cutting API latency by **50%**.
- Built a **roles and permissions** API, integrated with the UI, that secures all **7+** backend services.
- Deployed 7+ backend services and UI frontends (React, Next.js, FastAPI); added **Liquibase** migrations to the auth MySQL database, cutting manual schema-change time by **20%**; ran three **gRPC** proofs of concept for moving services off REST.

VENTURES & TEACHING

Chukta · Solo builder, side project · getchukta.in

Oct 2026 - Present · Android beta

- A **free, ad-free Splitwise alternative** for friends in India: **one-tap UPI settle-up**, receipt scanning with item-by-item splits, offline use, Splitwise import, WhatsApp reminders and pay-me links. **Android beta live** (Oct 2026); iPhone web app at app.getchukta.in.
- One **Expo / React Native** codebase (TypeScript, NativeWind) for the Android app and the iPhone PWA, on **Supabase**: Postgres with row-level security, Auth, Edge Functions and pg_cron.
- Money in **integer paise**, never floats. Every split is recomputed on the server by a PL/pgSQL port of the TypeScript split logic; a differential test compares the two on **2,000** random groups to the paisa, alongside pgTAP, Vitest and fast-check tests.
- **On-device OCR** (ML Kit, Apple Vision, Tesseract.js on web) with an optional AI check verified to the paisa; a **replay-safe offline outbox**; a Splitwise import that rolls back unless every balance matches the file.

Markdrop · Founder, side venture · markdrop.in

Mar 2026 - Present

- Publish markdown and AI docs with no account, or upload HTML, PDFs, sheets, Word docs, video and zipped sites: **one link that renders in any browser**.
- Web app, **VS Code extension** and a **CLI on Homebrew and Scoop**; TypeScript and Next.js, designed and built solo.

System Design Mastery · free course · himanshukukreja.in/learn

- Free course for backend engineers with **1,000+** quick-check questions, plus long-form **engineering case files**.

OPEN SOURCE & PROJECTS

ios-web-streamer WebRTC · Stream and control a real iOS device from a browser.

obscura TypeScript · HLS · Private video: self-hostable HLS from S3 with encrypted segments.

waygate Go · Self-hosted tunnel for exposing local services through your own infra.

codeflow-analyzer npm · CLI that statically analyzes React and Next.js codebases into behavior reports.

- **LogIngestor** Nov 2023 · Scalable log ingestion on FastAPI, RabbitMQ, MongoDB and Elasticsearch, with JWT auth, asyncio concurrency and rate limiting; Docker on AWS ECS and ECR.
- **AI-powered smart glasses** Oct – Dec 2022 · team of 5 · For visually impaired people: face recognition, image captioning, currency and traffic/object detection on a NodeMCU ESP8266 with Python and TensorFlow.
- **SmartAgri Solutions** Mar 2022 · Crop recommendations with a Random Forest at **96%** accuracy, fertilizer prediction and TensorFlow plant-disease detection (up to 80%); Flask on GCP.

SKILLS

LANGUAGES	Python, TypeScript, JavaScript, Go, SQL, Bash, C++
BACKEND	FastAPI, Flask, Django, Node.js, Bun, microservices, REST, gRPC, WebSockets, JWT and social auth
DATA	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, RabbitMQ, Supabase, Liquibase
CLOUD & INFRA	AWS (ECS, EFS, ECR, Lambda), GCP Cloud Run, Docker, CI/CD, GitHub Actions, Nginx, Terraform, Ansible, Linux
REALTIME & DEVICES	WebRTC, Android (ADB) and iOS (simctl, idb) device control, secure TCP tunnels, HLS video, noVNC
AI & AGENTS	LLM agents, MCP, Playwright automation, on-device OCR, TensorFlow
MOBILE & FRONTEND	React Native, Expo, React, Next.js, Electron, Tailwind / NativeWind, PWAs, CLIs and VS Code extensions
TESTING	Vitest, fast-check (property tests), pgTAP, Playwright

EDUCATION

B.E. Computer Science · Thapar Institute of Engineering & Technology

Aug 2019 - Jul 2023 · CGPA **9.04**/10

COURSES

Using Databases with Python · University of Michigan · Sep – Dec 2022

Web Application Technologies and Django · University of Michigan · May – Jul 2020